Module 26 – Sales Forecasting Case Study

Objective

In this module, you will work on a **Sales Forecasting** case study to apply time series or regression-based predictive modeling techniques. This exercise will help you understand how to analyze sales data, prepare it for forecasting, and make predictions using appropriate models.

Reference

Watch the **Module 26 video** carefully. Follow the explanation and steps provided in the video to complete your case study.

Dataset

You can use a **Sales Forecasting dataset** available on **Kaggle**.
Search for “**Sales Forecasting Dataset**” or any relevant dataset that includes sales data over time (for example, daily, weekly, or monthly sales data).

Task Instructions

1. Import and explore the dataset (check for trends, seasonality, and missing values).
2. Perform data preprocessing:
    - Handle missing values
    - Convert date columns into appropriate datetime format
    - Extract relevant features (month, day, year, etc.)
3. Split the dataset into **training** and **testing** sets.
4. Apply a suitable forecasting model (e.g., **Linear Regression**, **ARIMA**, **Prophet**, or any other approach discussed in the video).
5. Evaluate the model performance using relevant metrics such as **RMSE**, **MAE**, or **R^2 score**.
6. Summarize your observations and findings in a brief analysis section.

Submission Requirements

- Submit your `.ipynb` file (Jupyter Notebook).
- Upload your notebook to **Google Colab** and share the **Colab link**.
- Make sure your Colab link has **“Anyone with the link can view”** access.